

Quantum Error Correction

Quantum Error Mitigation

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Preliminaries and Motivation

Decoherence

- Loss of coherence due to 'unwanted' interactions with a bath/environment
 $\Rightarrow$ **Noise!**
- Mathematically, this gives rise to **c**ompletely **p**ositive, **t**race-**p**reserving (CPTP) maps.

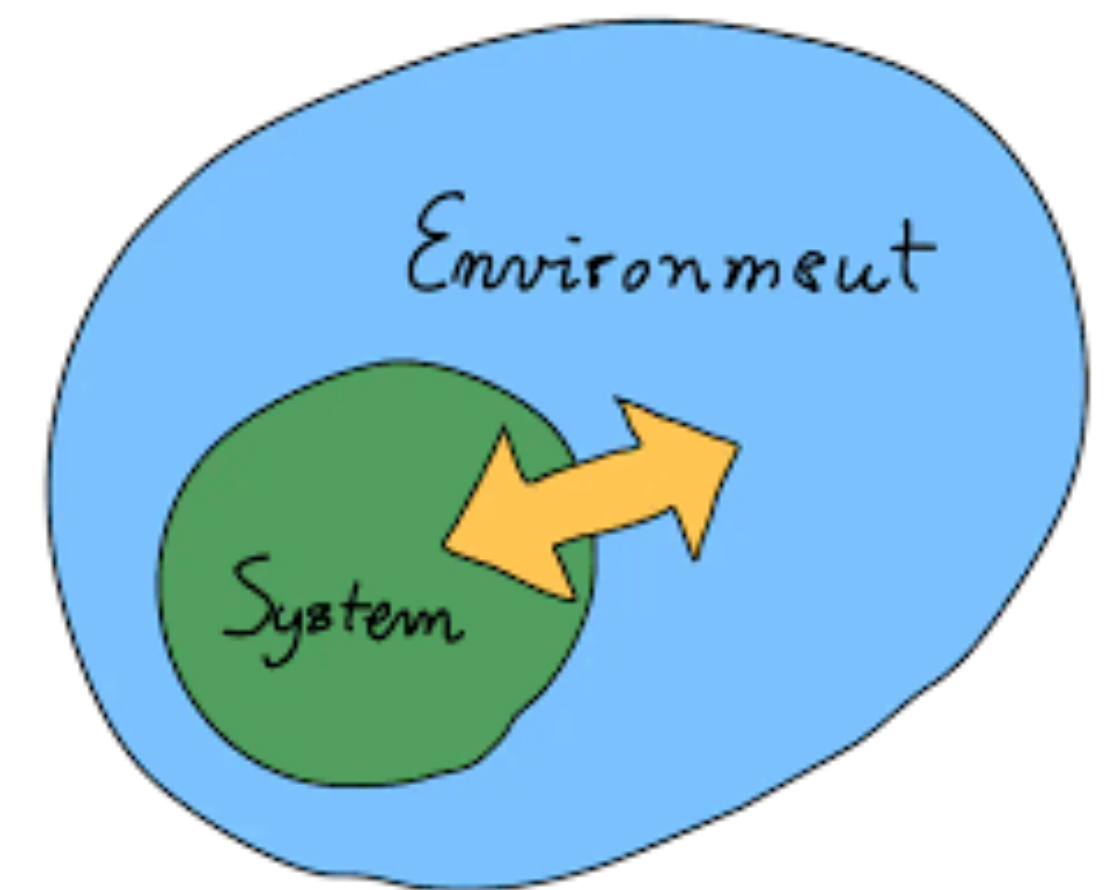
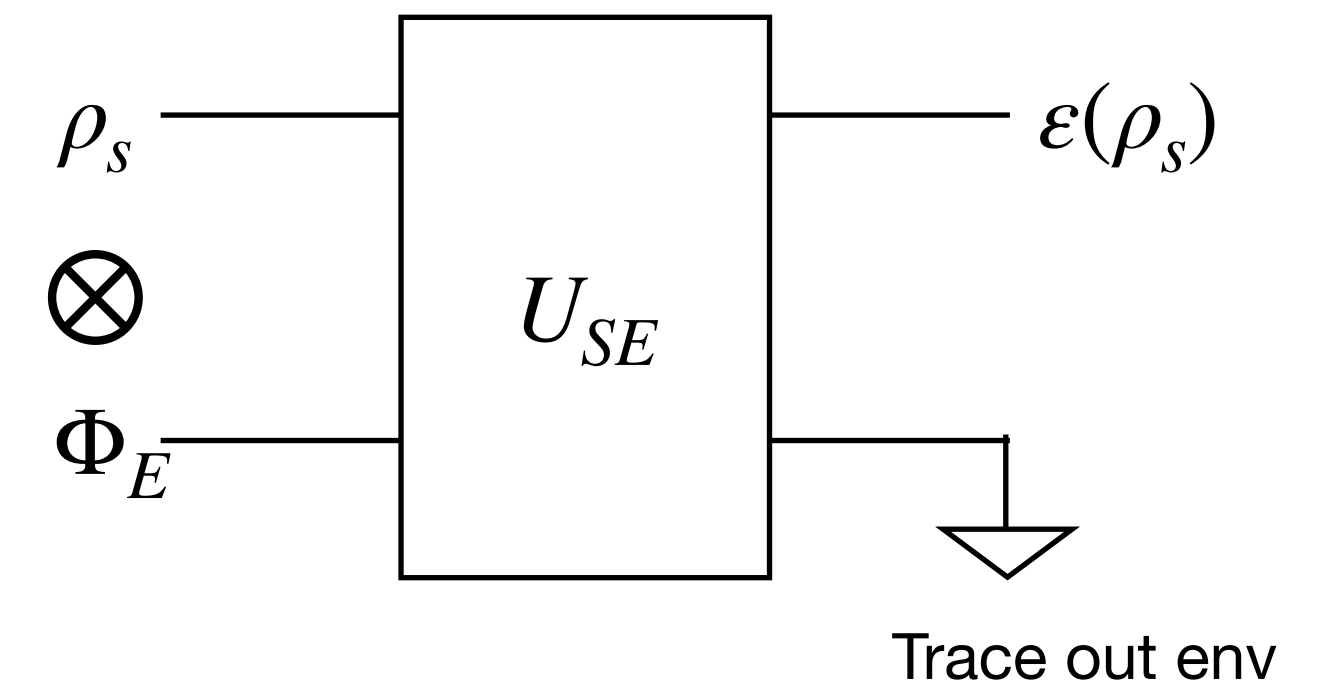
$$\mathcal{E}(\rho_S) = \text{tr}_E[U_{SE}(\rho_S \otimes \Phi_E)U_{SE}^\dagger].$$

Any physical process $\mathcal{E}$ on $\mathcal{H}_S$ must be,

- (i) **Completely positive (CP)**: $\mathcal{E}(\rho) > 0$, for all $\rho > 0 \in \mathcal{H}_S$.
And, $(\mathcal{E} \otimes \mathbb{I})$ is a positive map for any possible extension $\mathcal{H}_S \otimes \mathcal{H}_R$.
 $\Leftrightarrow$ **Choi-Kraus-Sudarshan operator-sum** representation:

$$\mathcal{E} \sim \{E_i\}_{i=1}^N : \mathcal{E}(\rho) = \sum_{i=1}^N E_i \rho E_i^\dagger.$$

- (ii) **Trace-Preserving** (TP) map : $\text{tr}[\mathcal{E}(\rho)] = 1 \Leftrightarrow \sum_i E_i^\dagger E_i = I_S$.



[Nielsen and Chuang, QCQI]

Preliminaries and Motivation

Amplitude Damping Channel

Effects due to loss of energy dissipation in a two-level system

Decay of the probability amplitudes from its excited state to ground state

[Nielsen and Chuang, QCQI]

Kraus operators

$$\mathcal{E}^{AD} = \left\{ E_0^{AD} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix} ; E_1^{AD} = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} \right\}$$

Probability of a transition from the excited state to the ground state

$$\gamma = 1 - e^{-\Gamma_1 t}$$

Inverse of relaxation time T_1

$$\Gamma_1 = 1/T_1$$

Phase Damping Channel

Loss of relative phase information between the energy eigenstates

Phase damping channel = Phase flip channel

[Nielsen and Chuang, QCQI]

Kraus operators

$$\mathcal{E}^{PD} = \left\{ E_0^{PD} = \begin{pmatrix} \sqrt{p} & 0 \\ 0 & \sqrt{p} \end{pmatrix} ; E_1^{PD} = \begin{pmatrix} \sqrt{1-p} & 0 \\ 0 & -\sqrt{1-p} \end{pmatrix} \right\}$$

Probability of not a phase flip

$$p = \frac{1}{2}(1 + e^{-\Gamma_\phi t})$$

Inverse of relaxation time T_ϕ

$$\Gamma_\phi = 1/T_\phi$$

Consider an arbitrary single-qubit state, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, Let $\rho = |\psi\rangle\langle\psi|$

$$\rho = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \xrightarrow{\text{noise}} \rho_{\text{noisy}} = \mathcal{E}^{AD} \circ \mathcal{E}^{PD}(\rho) = \mathcal{E}^{PD} \circ \mathcal{E}^{AD}(\rho) = \begin{pmatrix} 1 + (|\alpha|^2 - 1)e^{-\Gamma_1 t} & \alpha\beta^* e^{-\frac{\Gamma_1}{2}t} e^{-\Gamma_\phi t} \\ \alpha^*\beta e^{-\frac{\Gamma_1}{2}t} e^{-\Gamma_\phi t} & |\beta|^2 e^{-\Gamma_1 t} \end{pmatrix}$$

Why Quantum Error Correction?

*“**Noisy Intermediate-Scale Quantum (NISQ)** technology will be available in the near future. Quantum computers with **50-100** qubits may be able to perform tasks which surpass the capabilities of today’s classical digital computers, but noise in quantum gates will limit the size of quantum circuits that can be executed reliably.....*

Quantum Error Correction (is) our basis for thinking that quantum computers are scalable to large devices solving hard problems.”

– John Preskill, Quantum Computing in the NISQ era and beyond.

Quantum 2, 79 (2018).

Goals of QEC

- Modelling noise (*decoherence*) in physical systems.
- Develop protocols to preserve quantum states with high *fidelity*, under reasonable assumptions about the noise.
- Example: Qubit subject to *amplitude damping* noise (spontaneous emission).

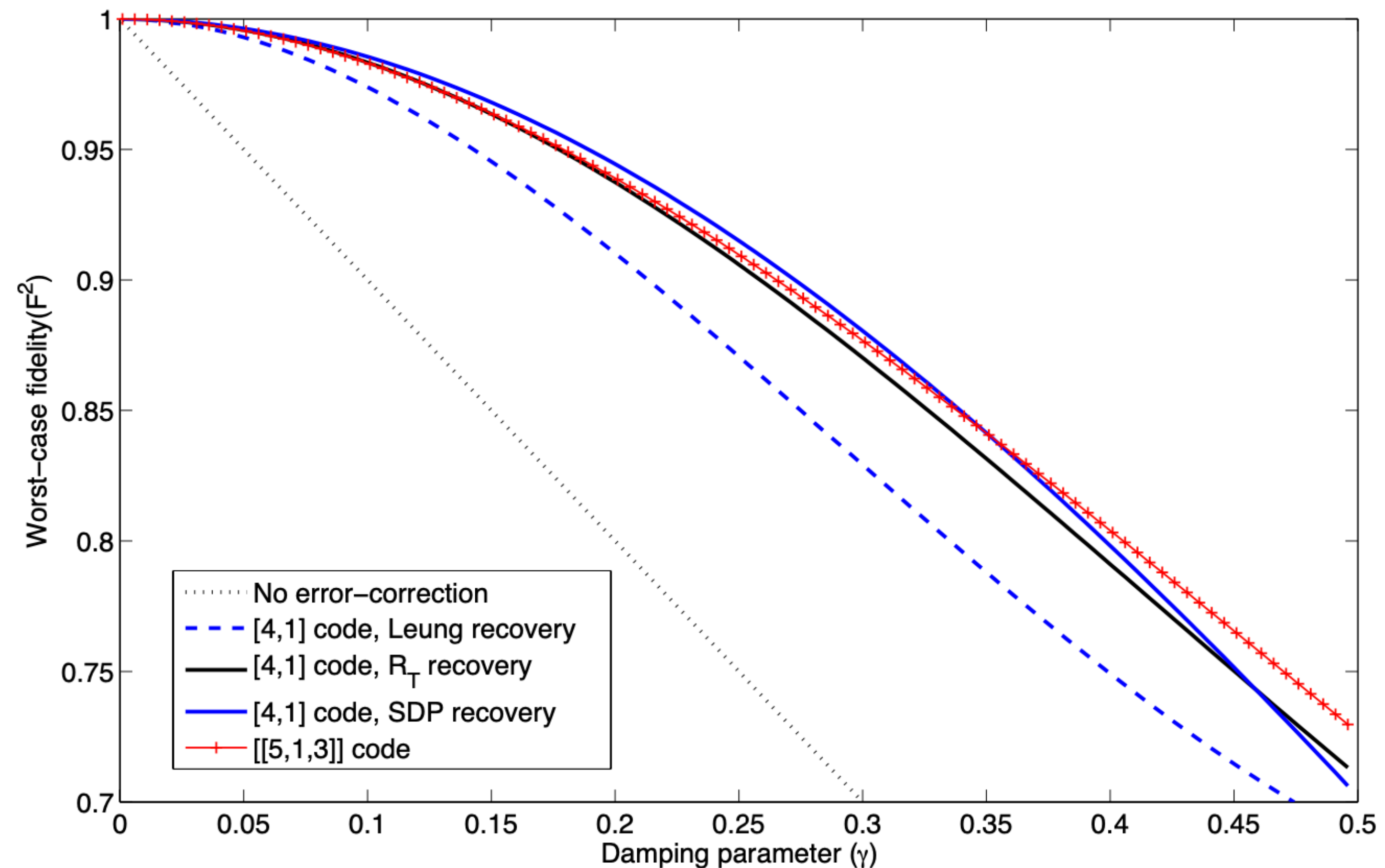
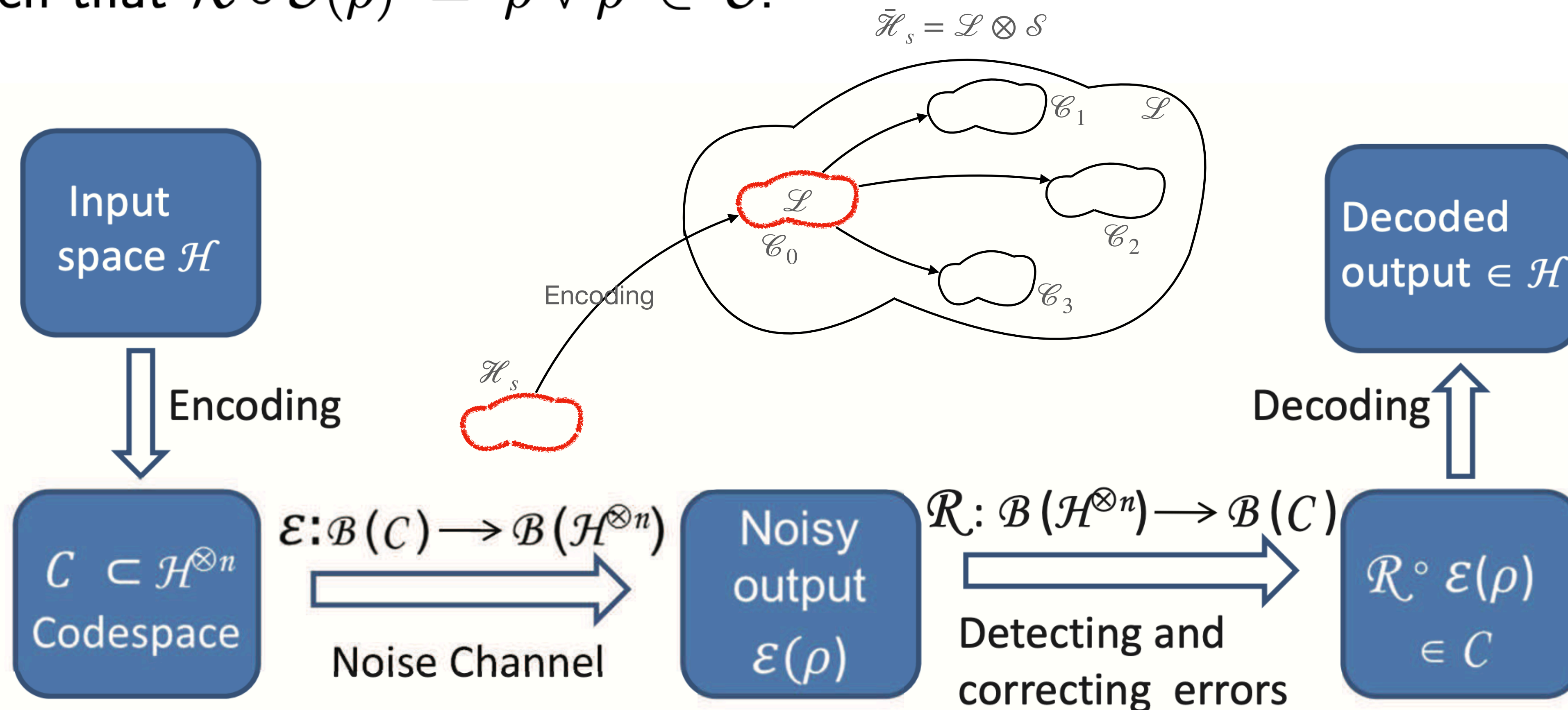


Figure: Fidelity vs Noise Strength for different QEC schemes under amplitude damping noise

Quantum Error Correction

- **Codespace:** d -dim subspace $\mathcal{C}$ of $\mathcal{H}^{\otimes n}$, encoding a qudit.
- **Recovery map:** $\mathcal{R}$ is any trace preserving CP map from $\mathcal{H}^{\otimes n}$ to $\mathcal{C}$.
- Action of a channel $\mathcal{E}$ is **correctible** on a space $\mathcal{C}$ if $\exists$ a quantum channel $\mathcal{R}$ such that $\mathcal{R} \circ \mathcal{E}(\rho) = \rho \forall \rho \in \mathcal{C}$.



Conditions for QEC

- 3-qubit code: $|0\rangle_L = |000\rangle$; $|1\rangle_L = |111\rangle$.
Corrects for single-qubit bit-flip errors.
- Given a noise channel $\mathcal{E} \sim \{E_i\}_{i=1}^N$, what is a good $\mathcal{C}$ to encode information in?

Let P be the projector onto codespace $\mathcal{C}$. A CPTP recovery map $\mathcal{R}_{\text{perf}}$ such that $\mathcal{R}_{\text{perf}} \circ \mathcal{E}(\rho) = \rho$ exists iff

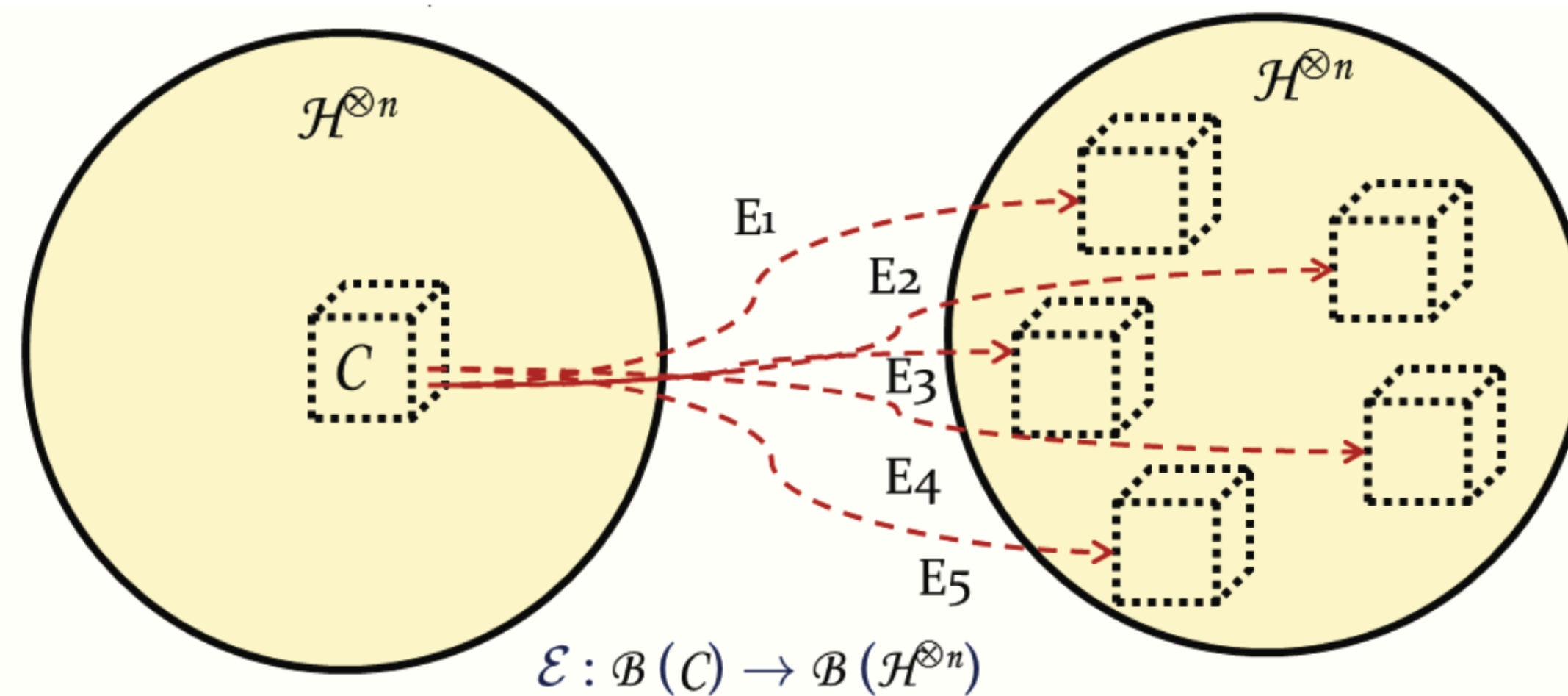
$$PE_i^\dagger E_j P = \alpha_{ij} P,$$

for some Hermitian matrix α of complex numbers.¹

- ▶ Easily verifiable, once P is given.
- ▶ **Linear** - any channel whose operator elements are linear combinations of $\{E_i\}$ is also correctible. For correcting single qubit errors, sufficient to check for the “Pauli errors” !

¹E.Knill and R.Laflamme, Phys.Rev. A **55**, 900 (1997).

Error operators map the codespace to **mutually orthogonal** subspaces of $(\mathcal{H})^{\otimes n}$.



- Recovery map $\mathcal{R}_{\text{Perf}} : \{R_k = PU_k^\dagger\}$.
- **Linearity** of QEC condition, and, assumption of **independent errors**
 $\Rightarrow$ The shortest perfect QEC code to correct arbitrary single qubit errors requires 5 qubits (**Five-qubit code**^{2,3})

²Bennet *et al.*, Phys.Rev.A **54** 3824 (1996)

³Laflamme *et al.*, Phys. Rev. Lett. **77**, 198 (1996)

Worst case Fidelity - Figure of merit

- Standard measure of closeness between two quantum states ρ, σ is the *fidelity*,

$$F(\rho, \sigma) = \text{tr} \sqrt{\rho^{1/2} \sigma \rho^{1/2}}.$$

When $\rho = |\psi\rangle\langle\psi|$, $F(|\psi\rangle, \sigma) = \sqrt{\langle\psi|\sigma|\psi\rangle}$.

- Worst-case fidelity:** For a codespace $\mathcal{C}$, under the action of the noise channel $\mathcal{E}$ and recovery $\mathcal{R}$,

$$F_{\min}[\mathcal{C}, \mathcal{R} \circ \mathcal{E}] = \min_{|\psi\rangle \in \mathcal{C}} F[|\psi\rangle, \mathcal{R} \circ \mathcal{E}(|\psi\rangle\langle\psi|)].$$

Suffices to minimize over pure states, since F is jointly concave in its arguments.

Classical Error Correction

- ❖ A single bit-flip error for classical data.

$$b \mapsto \begin{cases} b & \text{with probability } 1 - p \\ \neg b & \text{with probability } p. \end{cases}$$

- ❖ The classical repetition code:

encoding

$0 \rightarrow 000$

$1 \rightarrow 111$

transmission (3 bits)

$\longrightarrow$

decoding

$abc \rightarrow \text{majority}(a, b, c)$

If there is a single bit flip error, we can correct the state by choosing the majority of the three bits, e.g., $010 \rightarrow 0$. When errors are rare, one error is more likely than two.

- ❖ When can this help reduce errors?

$$\Pr[\text{correct transmission}] = (1 - p)^3 + 3(1 - p)^2 p = 1 - (3p^2 - 2p^3)$$

Quantum Error Correction

- ❖ Let us start with one of the simplest single-qubit error:

Bit Flip X: $X|0\rangle = |1\rangle, X|1\rangle = |0\rangle$

- ❖ Encoding:

$$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$$

- ❖ There is no device that will copy an unknown quantum state.

Superposition: $\alpha|0\rangle + \beta|1\rangle$

$$\alpha|0\rangle + \beta|1\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)^{\otimes 3} \quad \text{X}$$

Quantum Error Correction: Encoding

- ❖ But you can have quantum fanout instead of cloning

$$\alpha |0\rangle + \beta |1\rangle \rightarrow \alpha |0\rangle |0\rangle |0\rangle + \beta |1\rangle |1\rangle |1\rangle$$
$$\neq (\alpha |0\rangle + \beta |1\rangle) (\alpha |0\rangle + \beta |1\rangle) (\alpha |0\rangle + \beta |1\rangle)$$

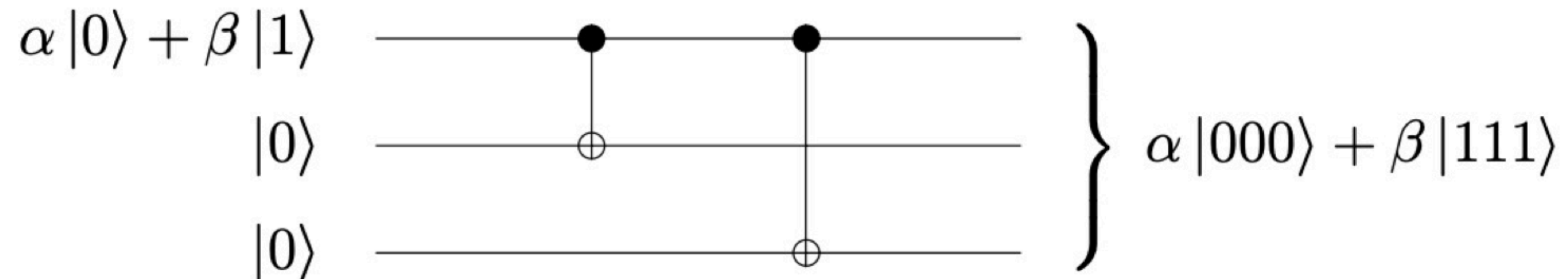
- ❖ A convenient way to write this encoding is:

$$|0\rangle \rightarrow |0_L\rangle \equiv |000\rangle$$

$$|1\rangle \rightarrow |1_L\rangle \equiv |111\rangle$$

We have repeated the state only in the computational basis

- ❖ This encoding could be performed by this simple circuit.



How to Detect Errors?

- ❖ Suppose we have done with the encoding for quantum repetition code:

$$\alpha |0\rangle + \beta |1\rangle \rightarrow \alpha |000\rangle + \beta |111\rangle$$

- ❖ Now in the transmission process, bit flip error (X) happens on 2nd qubit:

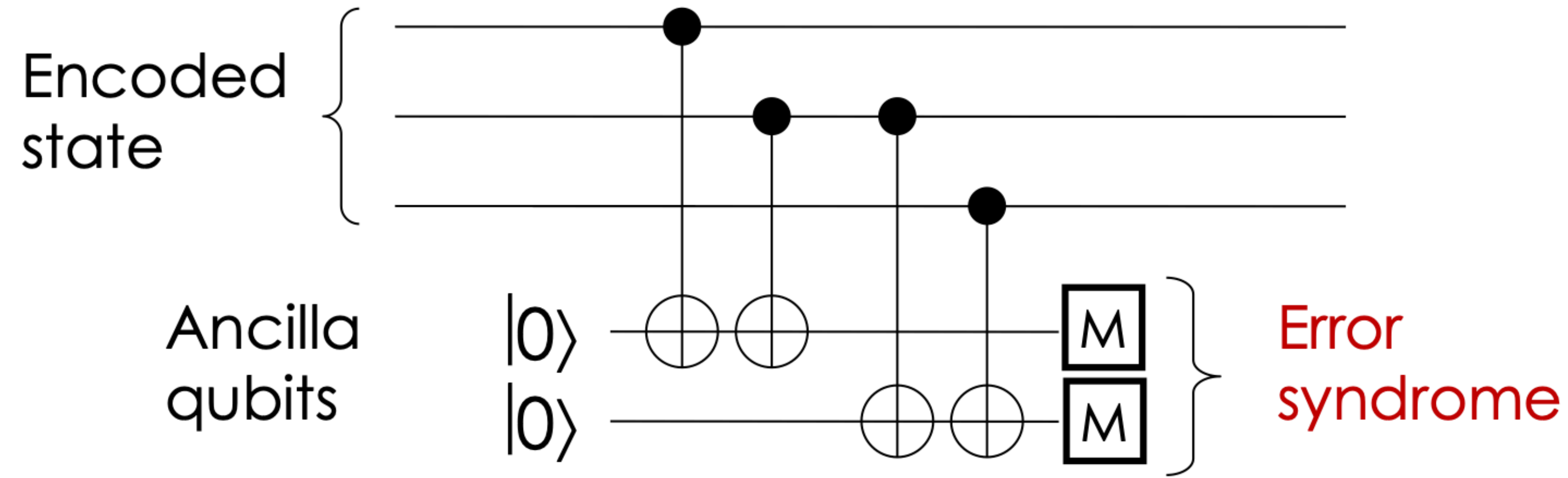
$$\alpha |010\rangle + \beta |101\rangle$$

- ❖ 2nd qubit is now **different** from 1st and 3rd.

- ❖ **Can we simply measure each of these three qubits or just one of them?**

Measure the Error, Not the Data

❖ Syndrome measurement:



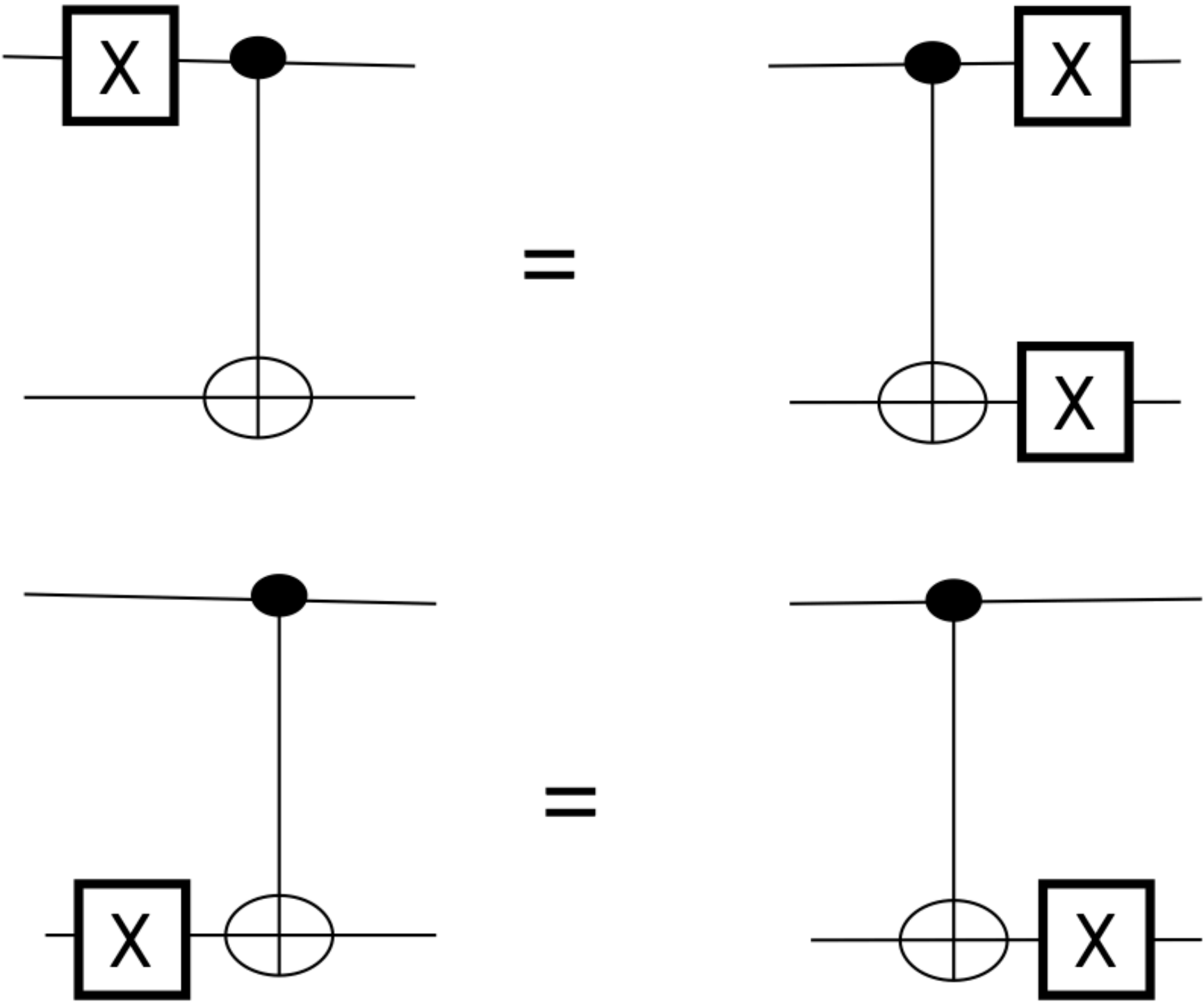
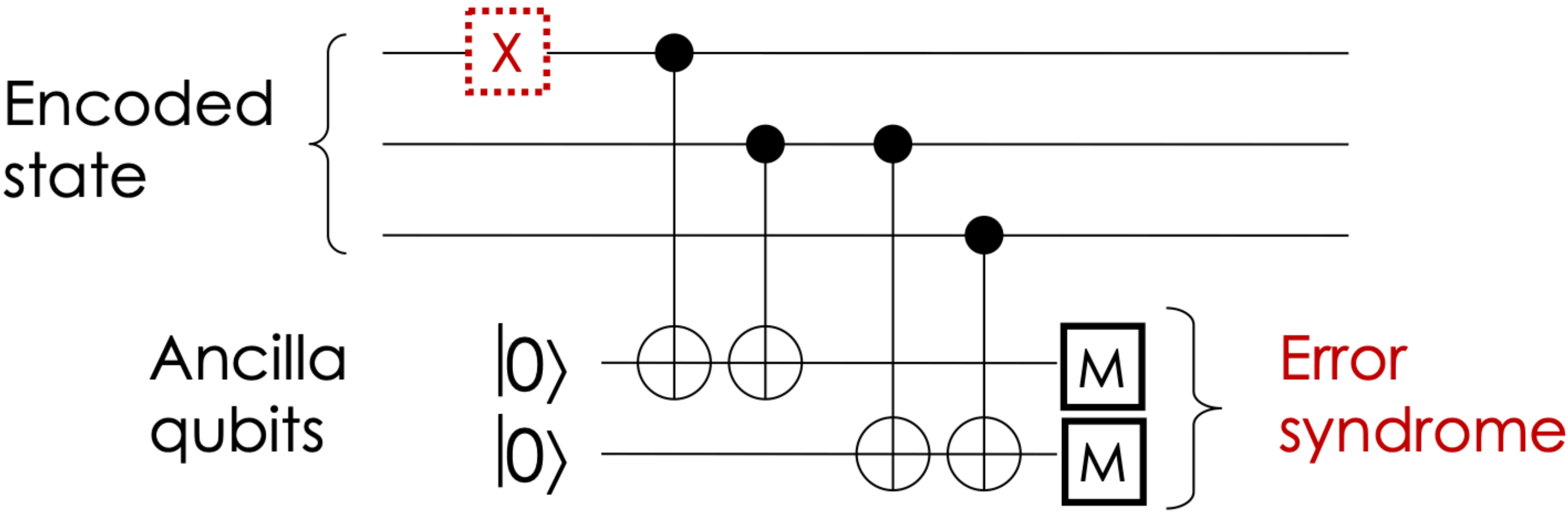
M stands for Z measurement

Let us try:

- $\alpha |100\rangle + \beta |011\rangle$
- $\alpha |010\rangle + \beta |101\rangle$
- $\alpha |001\rangle + \beta |110\rangle$

Measure the Error, Not the Data

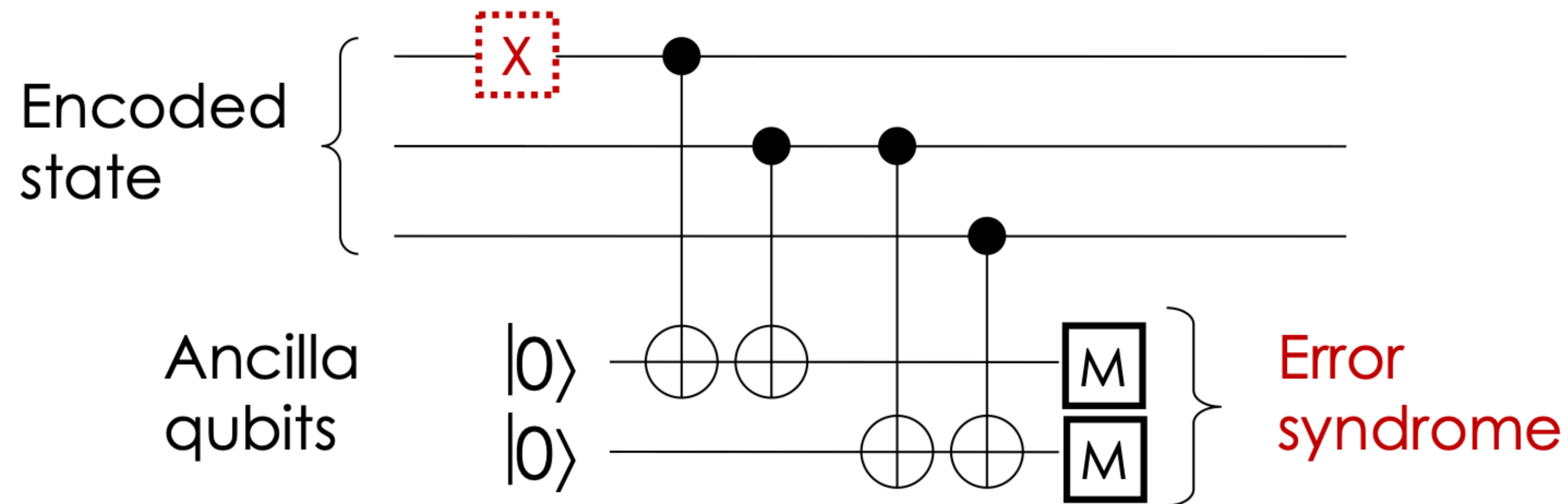
- ❖ A well-know pair of equivalent circuits:
- ❖ Can you check how X error propagate?



Summary of Syndrome Action and Error Reduction

❖ Measured syndrome action:

- 00 do nothing
- 10 apply X to first qubit
- 11 apply X to second qubit
- 01 apply X to third qubit



$$\text{Pr}[\text{correct transmission}] = (1 - p)^3 + 3(1 - p)^2 p = 1 - (3p^2 - 2p^3)$$

Quantum Error Correction

❖ Let us try another simple single-qubit error:

Phase error Z: $Z|0\rangle = |0\rangle, Z|1\rangle = -|1\rangle$

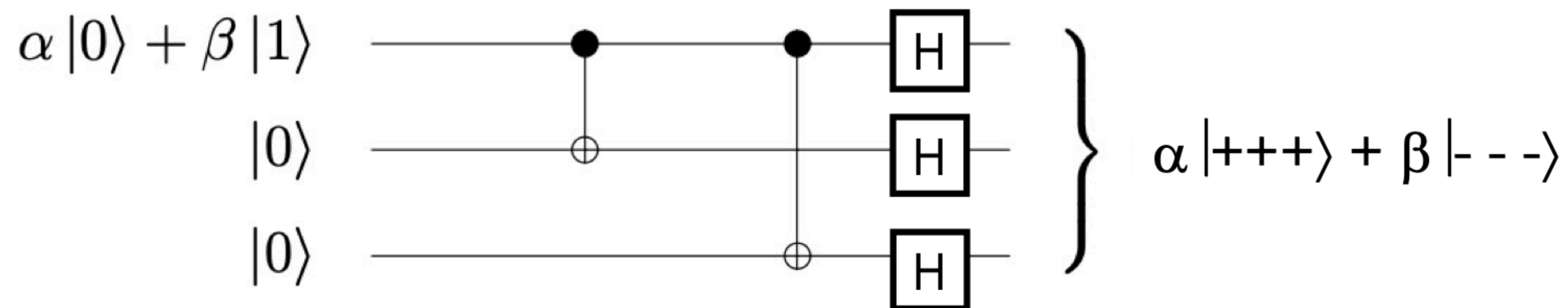
Look at another basis: $Z|+\rangle = |-\rangle, Z|-\rangle = |+\rangle$ (acts like bit flip)

❖ Different encoding: $\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|+++ \rangle + \beta|--- \rangle$

❖ A convenient way to write this encoding is:

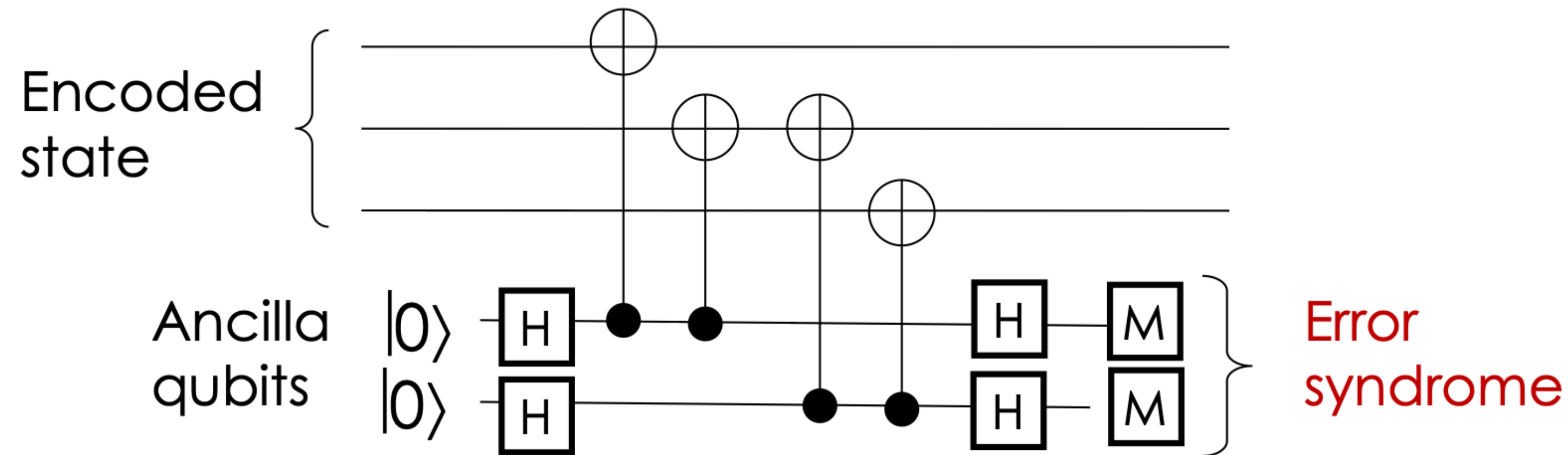
$$\begin{aligned} |0\rangle &\rightarrow |0_L\rangle \equiv |+++ \rangle \\ |1\rangle &\rightarrow |1_L\rangle \equiv |--- \rangle \end{aligned}$$

❖ This encoding could be performed by this simple circuit.



Measure the Error, Not the Data

❖ Syndrome measurement:



Try:

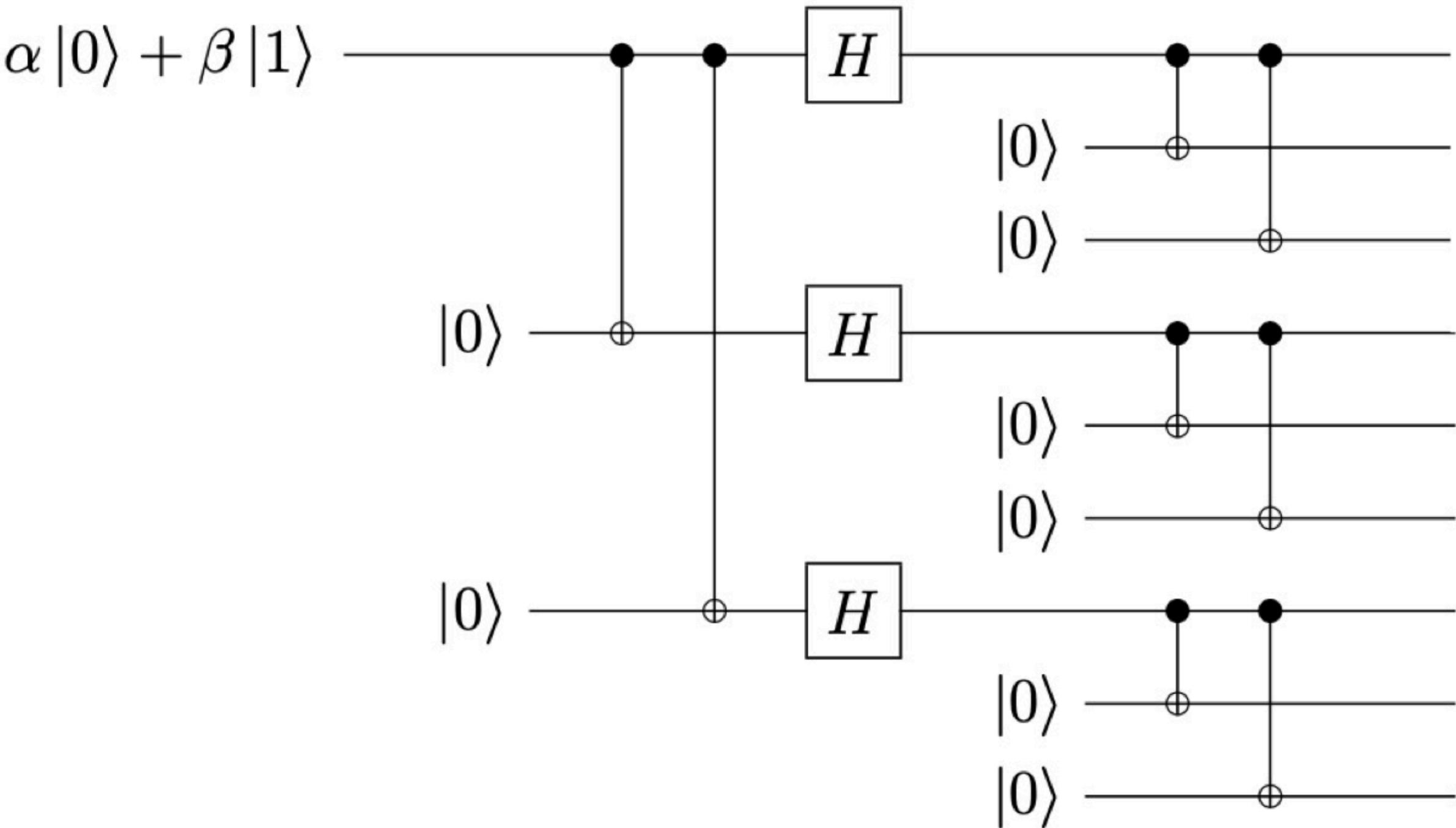
- $\alpha | - + + \rangle + \beta | + - - \rangle$
- $\alpha | + - + \rangle + \beta | - + - \rangle$
- $\alpha | + + - \rangle + \beta | - - + \rangle$

- ❖ 1st bit of error syndrome says whether the first two bits of the state are the same or different.
- ❖ 2nd bit of error syndrome says whether the second two bits of the state are the same or different.

Can we Correct both X and Z errors?

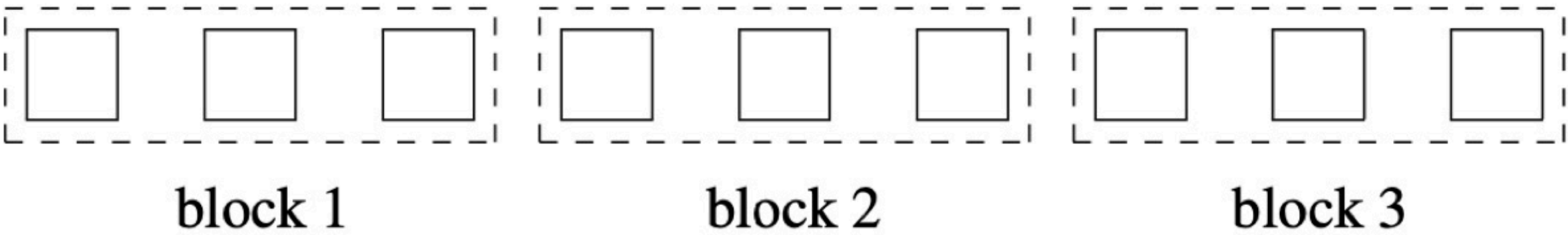
❖ Yes, Shor's 9 qubit code, which is first discovered by Peter Shor.

❖ Encoding circuit



$$\alpha \frac{(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)}{2\sqrt{2}} + \beta \frac{(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)}{2\sqrt{2}}.$$

❖ Each block represents an encoding of one of the three qubits of the state:



$$\alpha |+\rangle |+\rangle |+\rangle + \beta |-\rangle |-\rangle |-\rangle$$

Practice: Q&A

- ❖ The Shor 9 qubit code can correct X, Z, and XZ errors on any of the 9 qubits.
- ❖ Suppose the error **XZ** occurs on qubit number four. The encoded state becomes

$$\begin{aligned} & \alpha \frac{(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)}{2\sqrt{2}} \\ & + \beta \frac{(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)}{2\sqrt{2}} \end{aligned} \quad \longrightarrow \quad \begin{aligned} & \alpha \frac{(|000\rangle + |111\rangle)(|100\rangle - |011\rangle)(|000\rangle + |111\rangle)}{2\sqrt{2}} \\ & + \beta \frac{(|000\rangle - |111\rangle)(|100\rangle + |011\rangle)(|000\rangle - |111\rangle)}{2\sqrt{2}} \end{aligned}$$

- ❖ Eight Stabilizers:

$$\mathbf{Z_1Z_2}, \mathbf{Z_2Z_3}, \mathbf{Z_4Z_5}, \mathbf{Z_5Z_6}, \mathbf{Z_7Z_8}, \mathbf{Z_8Z_9}, \mathbf{X_1X_2X_3X_4X_5X_6}, \mathbf{X_4X_5X_6X_7X_8X_9}$$

- ❖ Which stabilizer syndrome measurement will give you the error information?

Z₄Z₅ This implies that a bit-flip error happened on qubit 4. Apply a X gate to qubit 4 will fix it.

$$\mathbf{X_1X_2X_3X_4X_5X_6}$$

This implies that a phase-flip happened somewhere in block 2. Applying a Z gate to any qubit in this block results will fix it.

Correcting All Single-Qubit Errors

- ❖ **Theorem:** If a quantum error-correcting code (QECC) corrects errors A and B , it also corrects $\alpha A + \beta B$.
- ❖ Any 2×2 matrix can be written as $\alpha I + \beta X + \gamma Y + \delta Z$.
- ❖ Any QECC that corrects the single-qubit errors X , Y , and Z (plus I) corrects every single-qubit error.